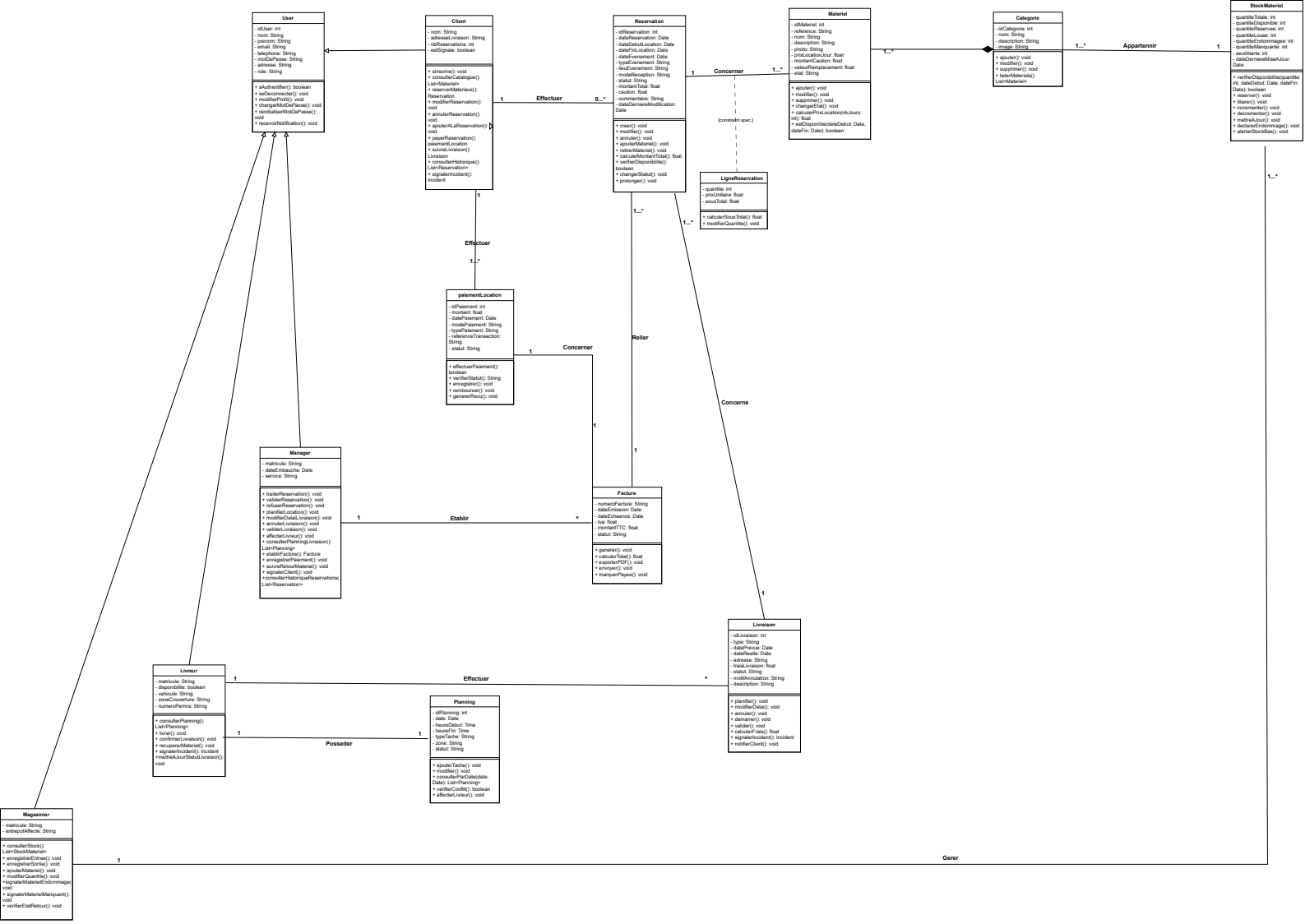

User
- idUser: int
- nom: String
- prenom: String
- email: String
- telephone: String
- motDePasse: String
- adresse: String
- role: String
+ sAuthentifier(): boolean
+ seDeconnecter(): void
+ modifierProfil(): void
+ changerMotDePasse(): void
+ reinitialiserMotDePasse(): void
+ recevoirNotification(): void
Client
- nom: String
- adresseLivraison: String
- ntbReservations: int
- estSignale: boolean
+ sInscrire(): void
+ consulterCatalogue(): List<Materiel>
+ reserverMateriaux(): Reservation
+ modifierReservation(): void
+ annulerReservation(): void
+ ajouterALaReservation(): void
+ payerReservation(): paiementLocation
+ suivreLivraison(): Livraison
+ consulterHistorique(): List<Reservation>
+ signalerIncident(): Incident
Effectuer
1
0...*
Reservation
- idReservation: int
- dateReservation: Date
- dateDebutLocation: Date
- dateFinLocation: Date
- dateEvenement: Date
- typeEvenement: String
- lieuEvenement: String
- modeReception: String
- statut: String
- montantTotal: float
- caution: float
- commentaire: String
- dateDerniereModification: Date
+ creer(): void
+ modifier(): void
+ annuler(): void
+ ajouterMateriel(): void
+ retirerMateriel(): void
+ calculerMontantTotal(): float
+ verifierDisponibilite(): boolean
+ changerStatut(): void
+ prolonger(): void
Concerner
1
1...*
Materiel
- idMateriel: int
- reference: String
- nom: String
- description: String
- photo: String
- prixLocationJour: float
- montantCaution: float
- valeurRemplacement: float
- etat: String
+ ajouter(): void
+ modifier(): void
+ supprimer(): void
+ changerEtat(): void
+ calculerPrixLocation(nbJours: int): float
+ estDisponible(dateDebut: Date, dateFin: Date): boolean
1...*
Categorie
- idCategorie: int
- nom: String
- description: String
- image: String
+ ajouter(): void
+ modifier(): void
+ supprimer(): void
+ listerMateriels(): List<Materiel>
1...*
Appartennir
1
StockMateriel
- quantiteTotale: int
- quantiteDisponible: int
- quantiteReservee: int
- quantiteLouee: int
- quantiteEndommagee: int
- quantiteManquante: int
- seuilAlerte: int
- dateDerniereMiseAJour: Date
+ verifierDisponibilite(quantite: int, dateDebut: Date, dateFin: Date): boolean
+ reserver(): void
+ liberer(): void
+ incrementer(): void
+ decrementer(): void
+ mettreAJour(): void
+ declarerEndommage(): void
+ alerterStockBas(): void
1...*
(constraint spec.)
LigneReservation
- quantite: int
- prixUnitaire: float
- sousTotal: float
+ calculerSousTotal(): float
+ modifierQuantite(): void
1...*
Effectuer
1...*
paiementLocation
- idPaiement: int
- montant: float
- datePaiement: Date
- modePaiement: String
- typePaiement: String
- referenceTransaction: String
- statut: String
+ effectuerPaiement(): boolean
+ verifierStatut(): String
+ enregistrer(): void
+ rembourser(): void
+ genererRecu(): void
Concerner
1
Relier
1...*
Concerne
1
Manager
- matricule: String
- dateEmbauche: Date
- service: String
+ traiterReservation(): void
+ validerReservation(): void
+ refuserReservation(): void
+ planifierLocation(): void
+ modifierDelaiLivraison(): void
+ annulerLivraison(): void
+ validerLivraison(): void
+ affecterLivreur(): void
+ consulterPlanningLivraison(): List<Planning>
+ etablirFacture(): Facture
+ enregistrerPaiement(): void
+ suivreRetourMateriel(): void
+ signalerClient(): void
+consulterHistoriqueReservations(List<Reservation>
1
Etablir
*
Facture
- numeroFacture: String
- dateEmission: Date
- dateEcheance: Date
- tva: float
- montantTTC: float
- statut: String
+ generer(): void
+ calculerTotal(): float
+ exporterPDF(): void
+ envoyer(): void
+ marquerPayee(): void
1
Livraison
- idLivraison: int
- type: String
- datePrevue: Date
- dateReelle: Date
- adresse: String
- fraisLivraison: float
- statut: String
- motifAnnulation: String
- description: String
+ planifier(): void
+ modifierDelai(): void
+ annuler(): void
+ demarrer(): void
+ valider(): void
+ calculerFrais(): float
+ signalerIncident(): Incident
+ notifierClient(): void
Livreur
- matricule: String
- disponibilite: boolean
- vehicule: String
- zoneCouverture: String
- numeroPermis: String
+ consulterPlanning(): List<Planning>
+ livrer(): void
+ confirmerLivraison(): void
+ recupererMateriel(): void
+ signalerIncident(): Incident
+mettreAJourStatutLivraison(): void
1
Effectuer
*
Planning
- idPlanning: int
- date: Date
- heureDebut: Time
- heureFin: Time
- typeTache: String
- zone: String
- statut: String
+ ajouterTache(): void
+ modifier(): void
+ consulterParDate(date: Date): List<Planning>
+ verifierConflit(): boolean
+ affecterLivreur(): void
1
Posseder
1
Magasinier
- matricule: String
- entrepotAffecte: String
+ consulterStock(): List<StockMateriel>
+ enregistrerEntree(): void
+ enregistrerSortie(): void
+ ajouterMateriel(): void
+ modifierQuantite(): void
+signalerMaterielEndommage(void
+ signalerMaterielManquant() void
+ verifierEtatRetour(): void
1
Gerer